

Mojtaba Jafarian Abyaneh

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SUMMARY

PhD candidate at Florida Atlantic University (I-SENSE Lab) developing Transformer-based deep learning models for multi-agent trajectory prediction of pedestrians, vehicles, and cyclists using LiDAR sensor data. First-author IEEE publication (IF: 6.007); second paper under review at *IEEE T-ITS*. Research funded by NSF (EEC-2133516) and NIH (R01AG068472).

EDUCATION

Florida Atlantic University

Ph.D. in Transportation and Environmental Engineering | GPA: 4.0/4.0

Boca Raton, FL

2024 – Present

- Focus: Transformer-based multi-agent trajectory prediction & LiDAR mobility modeling | Advisor: Dr. Jinwoo Jang

University of New Brunswick

M.Sc. in Civil Engineering

Fredericton, NB, Canada

2018 – 2020

EXPERIENCE

Doctoral Research Assistant

Aug. 2024 – Present

Florida Atlantic University (I-SENSE Lab)

Boca Raton, FL

- Developed **SIPT** (Spatial Interaction Pooling Transformer) for multi-agent trajectory prediction: **26.2% ADE improvement over Social LSTM**, 13.6% over Standard Transformer, on 64,000+ trajectories (COSMOS dataset); under review at *IEEE T-ITS* (w/ Columbia University).
- Developed Transformer model for real-world LiDAR-based trajectory prediction: **15.24%/13.88% ADE improvement over LSTM** (single-/multi-agent) on 12,390+ trajectories (West Palm Beach intersection); published in *IEEE OJ-ITS*, Vol. 7 (2026).
- Managed multi-sensor driving dataset (camera, LiDAR, IMU): 750 participants, 12 collection waves over 3 years; owned data pipeline, QA, and protocol compliance.

Graduate Research Mentor

Jan. 2026 – Present

VIP Program, Florida Atlantic University

Boca Raton, FL

- Building computer vision and smart sensor analytics pipeline for coastal transportation monitoring (NSF CS3); applying spatial data analysis and sensor fusion to multi-modal safety datasets.
- Mentoring REU students: FMM-based GPS map matching for driver behavior analysis (NIH-funded) and **QuASAR**, a watertight 3D city reconstruction pipeline for CFD-ready digital twins (NSF CS3).

TECHNICAL SKILLS

- **Programming:** Python, PyTorch, TensorFlow, Keras, Git, MATLAB, LaTeX
- **AI & ML:** Transformer Models, Spatial Graph Neural Networks, Computer Vision, LiDAR Point Cloud Processing, Multi-Agent Trajectory Prediction, Deep Learning, Scikit-learn
- **Data & GIS:** Pandas, NumPy, ArcGIS, Geopandas, Statistical Analysis

PUBLICATIONS

1. **Jafarian Abyaneh, M.** and Jang, J. (2026) "Transformer-Based Trajectory Prediction Using LiDAR Data for Situational Awareness in Complex Urban Environments." *IEEE Open Journal of Intelligent Transportation Systems*, Vol. 7, pp. 16–28. doi:10.1109/OJITS.2025.3640002 (IF: 6.007)
2. **Jafarian Abyaneh, M.**, Jang, J., Smyth, A. W., and Turkcan, M. K. (2026) "Spatial Interaction Pooling Transformer (SIPT) for Multi-Agent Trajectory Prediction in Urban Streetscapes." *IEEE Transactions on Intelligent Transportation Systems*. [Under Review]
3. **Jafarian Abyaneh, M.**, Jang, J., and Kaisar, E. I. (2025) "Spatial GCN for Predicting Bike-Sharing Origin–Destination Spatiotemporal Flows." *SSRN Working Paper* No. 5801889. [Preprint]

SELECTED PRESENTATIONS

- **CS3 Innovation Summit**, Columbia University, New York, NY (Feb. 2026) — SIPT multi-agent trajectory prediction; **Perfect Pitch Competition** (90-sec research pitch) & poster.
- **TRB 105th Annual Meeting**, Washington, DC (Jan. 2026) — Bike-sharing O-D spatiotemporal flow prediction; poster.